

Title

Hypertension severity modifies anticoagulation benefit in cryptogenic stroke with atrial cardiopathy: exploratory analysis from the ARCADIA trial

Abstract

Background: Disparate trial results in cryptogenic stroke may reflect heterogeneous underlying stroke mechanisms. We examined whether hypertension severity modifies anticoagulation efficacy. **Methods:** From 945 patients with cryptogenic stroke and atrial cardiopathy randomized to apixaban versus aspirin between 2018 and 2023, we analysed 945 with complete hypertension and imaging data. Hypertension with high-risk features (elevated systolic BP ≥ 160 mmHg and/or ventricular hypertrophy imaging) was present in 351 (37.1%) participants. **Results:** A significant treatment-by-hypertension interaction was observed ($p=0.024$). In 594 patients without high-risk hypertension, apixaban demonstrated superior efficacy (hazard ratio 0.43; 95% CI 0.22-0.85; absolute risk reduction 3.4% over median 1.6 years). Conversely, in 351 with hypertension high-risk features, no efficacy difference emerged (hazard ratio 1.68; 95% CI 0.78-3.62; $p=0.19$); absolute event rate difference 2.4% favoured aspirin. **Conclusions:** Advanced hypertension phenotypes may indicate stroke mechanisms resistant to anticoagulation benefit, possibly reflecting hypertensive arteriolosclerosis rather than cardioembolism. Trial design accounting for hypertension severity may better identify populations deriving anticoagulation benefit in cryptogenic stroke.